



UNIVERSITÀ
DI TORINO



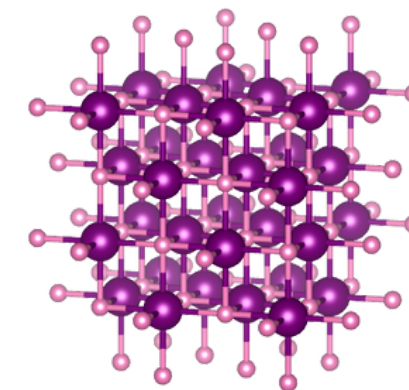
One-electron properties

Advanced tutorial session-Day 1

MSSC2025

What are we going to do today?

- * Band structure
- * Density of states
- * Additional/advanced keywords
- * CRYSTALClear notebooks as visualization tool



MgO crystal

- * 3D electrostatic potential maps for a MOF

How to compute one-electron properties with the CRYSTAL code?

One-electron properties and wave function analysis can be computed from the SCF wave function by running *properties*.

Data stored in “wavefunction file” (file.f9):

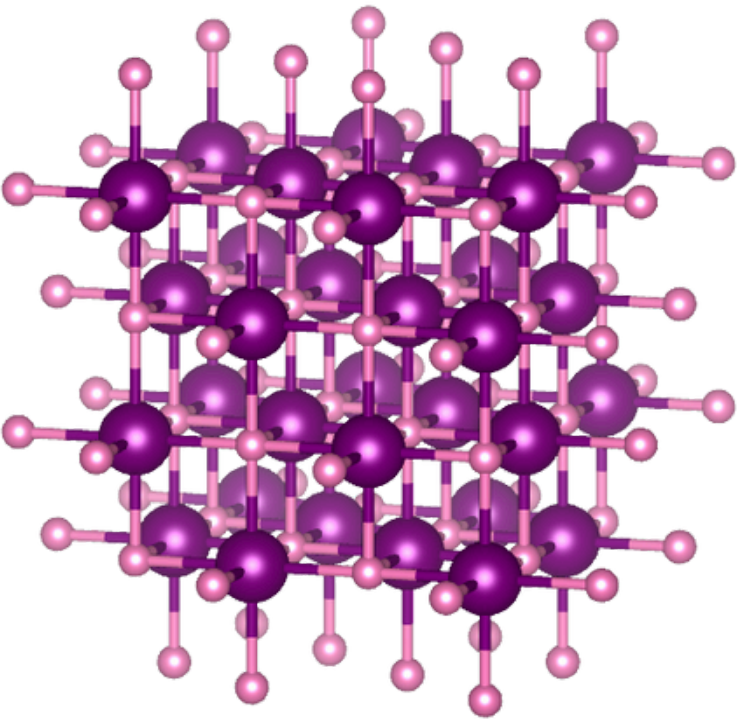
- Geometry, symmetry, crystal structure;
- basis set;
- reciprocal lattice k-point sampling information;
- irreducible Fock/KS matrix in direct space;
- irreducible density matrix in the direct space

MgO CRYSTAL reference input for obtaining the wavefunction

Input: *filename.d12*

MGO	Title
CRYSTAL	
0 0 0	
225	
4.21	
2	
12 0. 0. 0.	
8 0.5 0.5 0.5	
END	Geometry section
12 4	
0 0 8 2.0 1.0	From CRYSTAL basis set library:
68370.0 0.0002226	• Mg_8-511G_harrison_1994 • O_8-411_muscat_1999
...	
8 4	
0 0 8 2.1.	
8020.0 0.00108	
...	
99 0	
END	Basis set

DFT	
PBE0	
END	Method and functional
TOLINTEG	
8 8 8 8 16	
TOLDEE	
8	
SHRINK	
8 8	
END	Computational parameters



MgO CRYSTAL reference input for obtaining the wavefunction

Input: *filename.d12*

MGO	Title
CRYSTAL	
0 0 0	
225	
4.21	
2	
12 0. 0. 0.	
8 0.5 0.5 0.5	
END	Geometry section
12 4	
0 0 8 2.0 1.0	From CRYSTAL basis set library:
68370.0 0.0002226	• Mg_8-511G_harrison_1994
...	• O_8-411_muscat_1999
8 4	
0 0 8 2.1.	
8020.0 0.00108	
...	
99 0	
END	Basis set

DFT	
PBE0	
END	Method and functional
TOLINTEG	
8 8 8 8 16	
TOLDEE	
8	
SHRINK	
8 8	
END	Computational parameters

HOW TO RUN CRYSTAL:

runCRY23 *input filename*
(without .d12 extension)

MgO CRYSTAL reference input for obtaining the wavefunction

Input: *filename.d12*

MGO	Title
CRYSTAL	
0 0 0	
225	
4.21	
2	
12 0. 0. 0.	
8 0.5 0.5 0.5	
END	Geometry section
12 4	
0 0 8 2.0 1.0	From CRYSTAL basis set library:
68370.0 0.0002226	• Mg_8-511G_harrison_1994 • O_8-411_muscat_1999
...	
8 4	
0 0 8 2.1.	
8020.0 0.00108	
...	
99 0	
END	

DFT	
PBE0	
END	Method and functional
TOLINTEG	
8 8 8 8 16	
TOLDEE	
8	
SHRINK	
8 8	
END	Computational parameters

Ionic compound, insulator; experimental direct band gap= 7.8 eV

MgO CRYSTAL reference input for obtaining the wavefunction

Input: *filename.d12*

MGO	Title
CRYSTAL	
0 0 0	
225	
4.21	
2	
12 0. 0. 0.	
8 0.5 0.5 0.5	
END	Geometry section
12 4	
0 0 8 2.0 1.0	From CRYSTAL basis set library:
68370.0 0.0002226	• Mg_8-511G_harrison_1994 • O_8-411_muscat_1999
...	
8 4	
0 0 8 2.1.	
8020.0 0.00108	
...	
99 0	
END	

DFT	
PBE0	
END	Method and functional
TOLINTEG	
8 8 8 8 16	
TOLDEE	
8	
SHRINK	
8 8	
END	Computational parameters

Ionic compound, insulator; experimental direct band gap= 7.8 eV
Final computed band gap: 7.42 eV (direct)

MgO

From CRYSTAL wavefuntion to properties calculation: **BAND STRUCTURE**

Calculation of the band structure from density matrix is a key way to describe the electronic structure of a material, since it tells us if it acts like a conductor, an insulator, a semiconductor; i.e. we can verify/predict how the introduction of defect may influence the behaviour of the material, etc...

Even the topology of occupied manifold and first conduction band can provide us insights into a material's properties.

Input: *filename.d3*

BAND	Keyword
Path: G-X-W-L-G	Title
4 8 60 7 14 1 0	Parameters
0 0 0 4 0 4	
4 0 4 4 2 6	
4 2 6 4 4 4	
4 4 4 0 0 0	Path in reciprocal space
END	

MgO

From CRYSTAL wavefuntion to *properties* calculation: **BAND STRUCTURE**

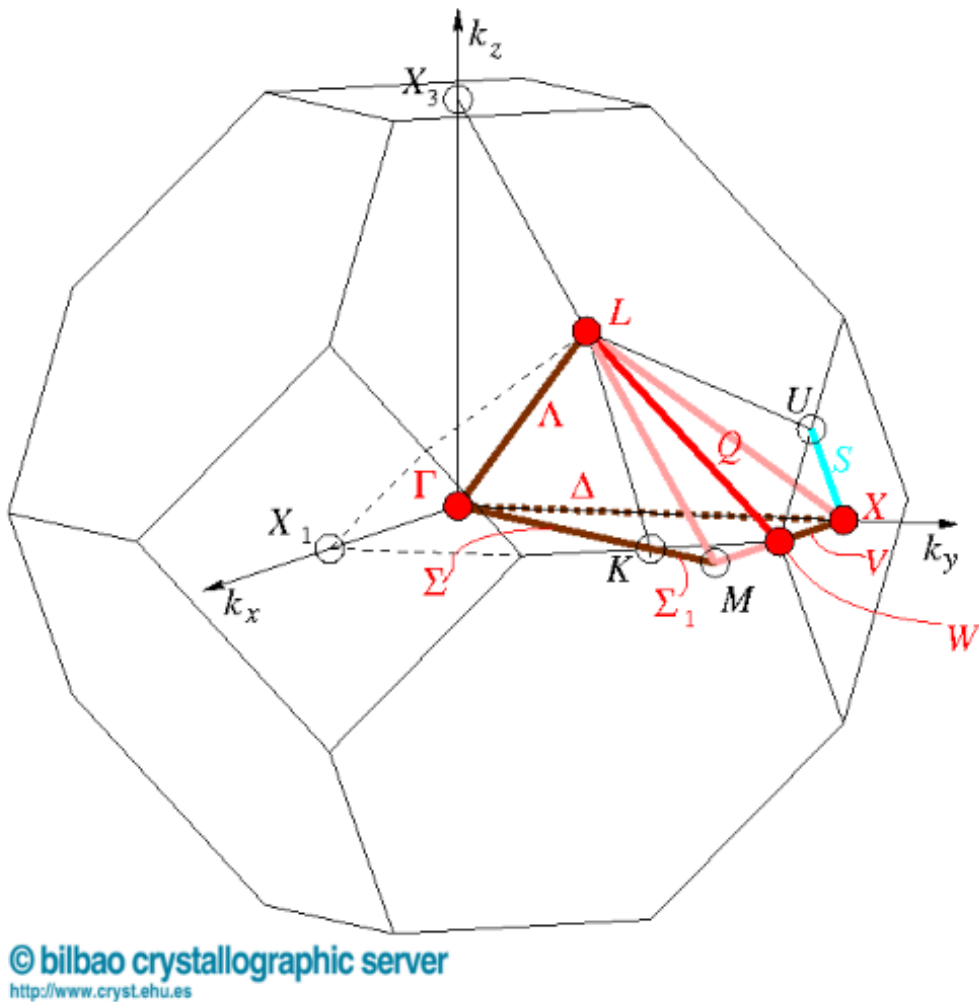
Calculation of the band structure from density matrix is a key way to describe the electronic structure of a material, since it tells us if it acts like a conductor, an insulator, a semiconductor; i.e. we can verify/predict how the introduction of defect may influence the behaviour of the material, etc...

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Input: *filename.d3*

BAND	Keyword
Path: G-X-W-L-G	Title
4 8 60 7 14 1 0	Parameters
0 0 0 4 0 4	
4 0 4 4 2 6	
4 2 6 4 4 4	
4 4 4 0 0 0	Path in reciprocal space *
END	

- * 1.BILBAO
CRYSTALLOGRAPHIC
SERVER
- 2.Space group symmetry
- 3.KVEC (primitive basis
coordinates)



MgO From CRYSTAL wavefunction to *properties* calculation: **BAND STRUCTURE**

Calculation of the band structure from density matrix is a key way to describe the electronic structure of a material, since it tells us if it acts like a conductor, an insulator, a semiconductor; i.e. we can verify/predict how the introduction of defect may influence the behaviour of the material, etc...

Even the topology of occupied manifold and first conduction band can provide us insights into a material's properties.

Input: *filename.d3*

BAND	Keyword
Path: G-X-W-L-G	Title
4 8 60 7 14 1 0	Parameters
0 0 0 4 0 4	
4 0 4 4 2 6	
4 2 6 4 4 4	
4 4 4 0 0 0	Path in reciprocal space
END	

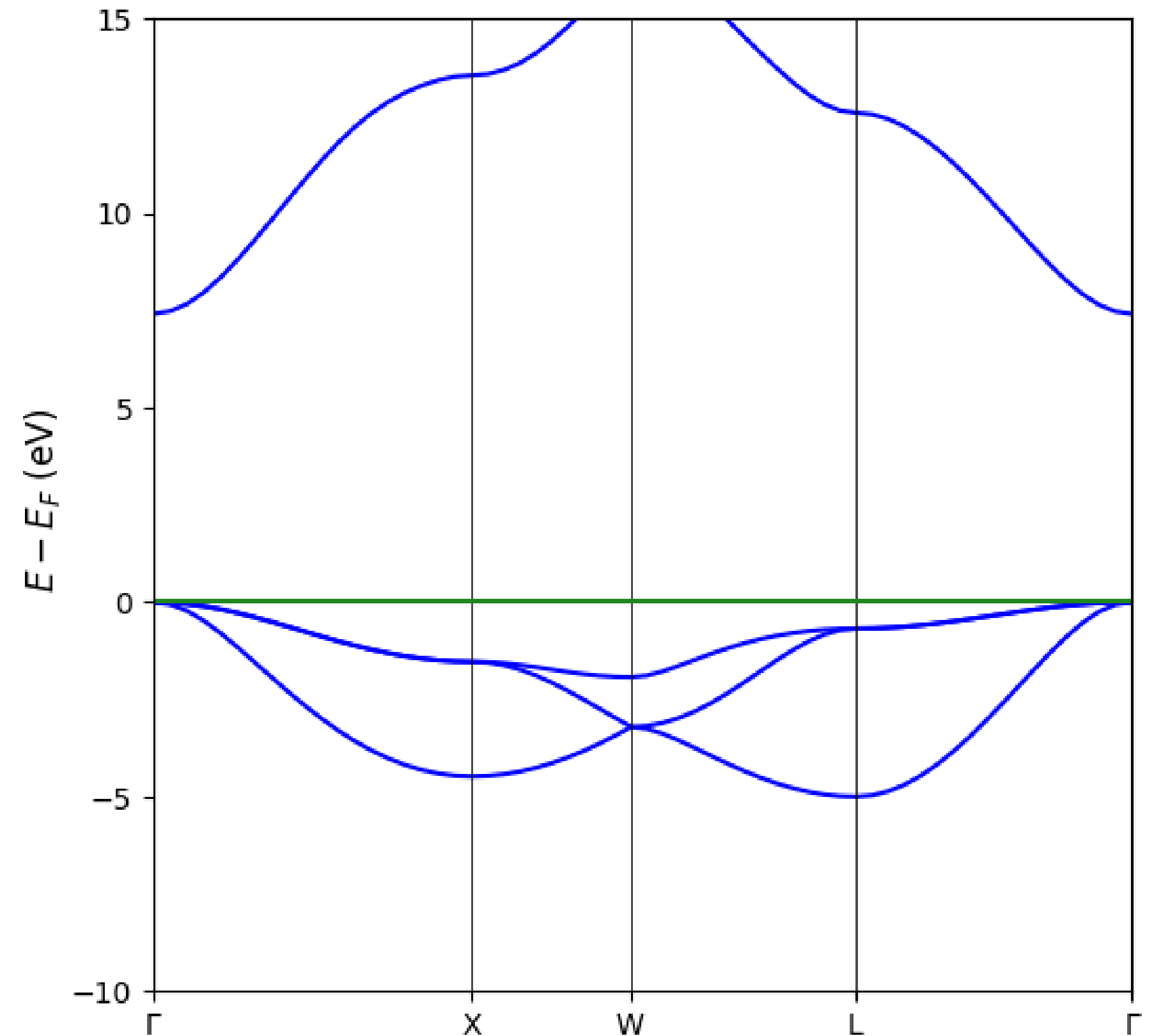
HOW TO RUN PROPERTIES:

(For all *properties* inputs)

runPROP23 *input filename wavefunction filename*
(without **.d3** and **.f9** extensions)

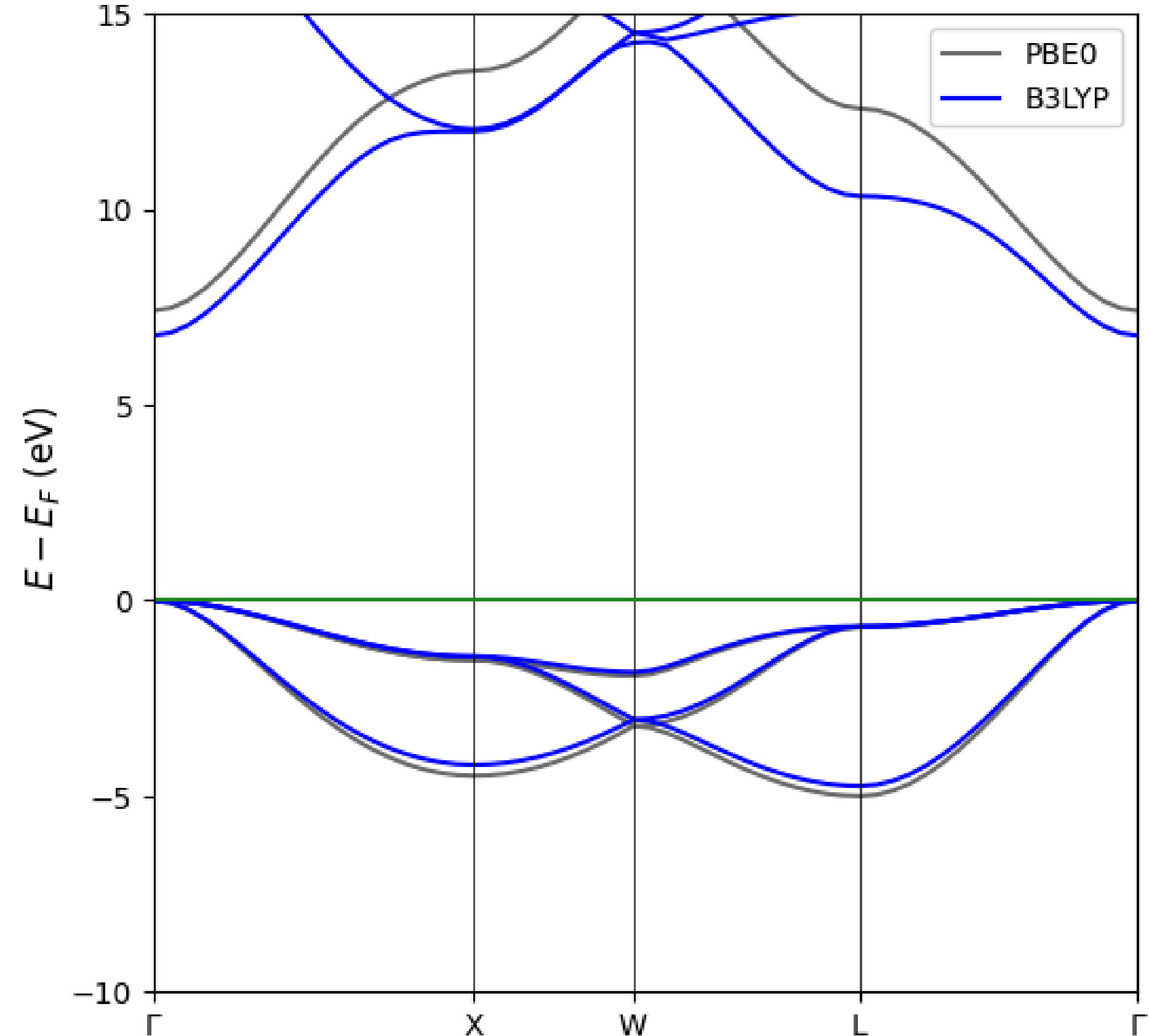
Calculation of the band structure from density matrix is a key way to describe the electronic structure of a material, since it tells us if it acts like a conductor, an insulator, a semiconductor; i.e. we can verify/predict how the introduction of defect may influence the behaviour of the material, etc...

Even the topology of occupied manifold and first conduction band can provide us insights into a material's properties.



Calculation of the band structure from density matrix is a key way to describe the electronic structure of a material, since it tells us if it acts like a conductor, an insulator, a semiconductor; i.e. we can verify/predict how the introduction of defect may influence the behaviour of the material, etc...

Even the topology of occupied manifold and first conduction band can provide us insights into a material's properties.



Calculation of density of states allows to obtain a chemical insight into the band structure, understanding which atomic species and even which atomic orbitals contribute the most to valence and conduction bands

Two different approaches:

Input: *filename.d3*

Projection on the total set
of AOs of a given atom

```
NEWK*
8 8
1 0
-----
DOSS                                Keyword
2 200 7 14 1 12 0                Parameters
-11
-1 2                                AO projection
END
```

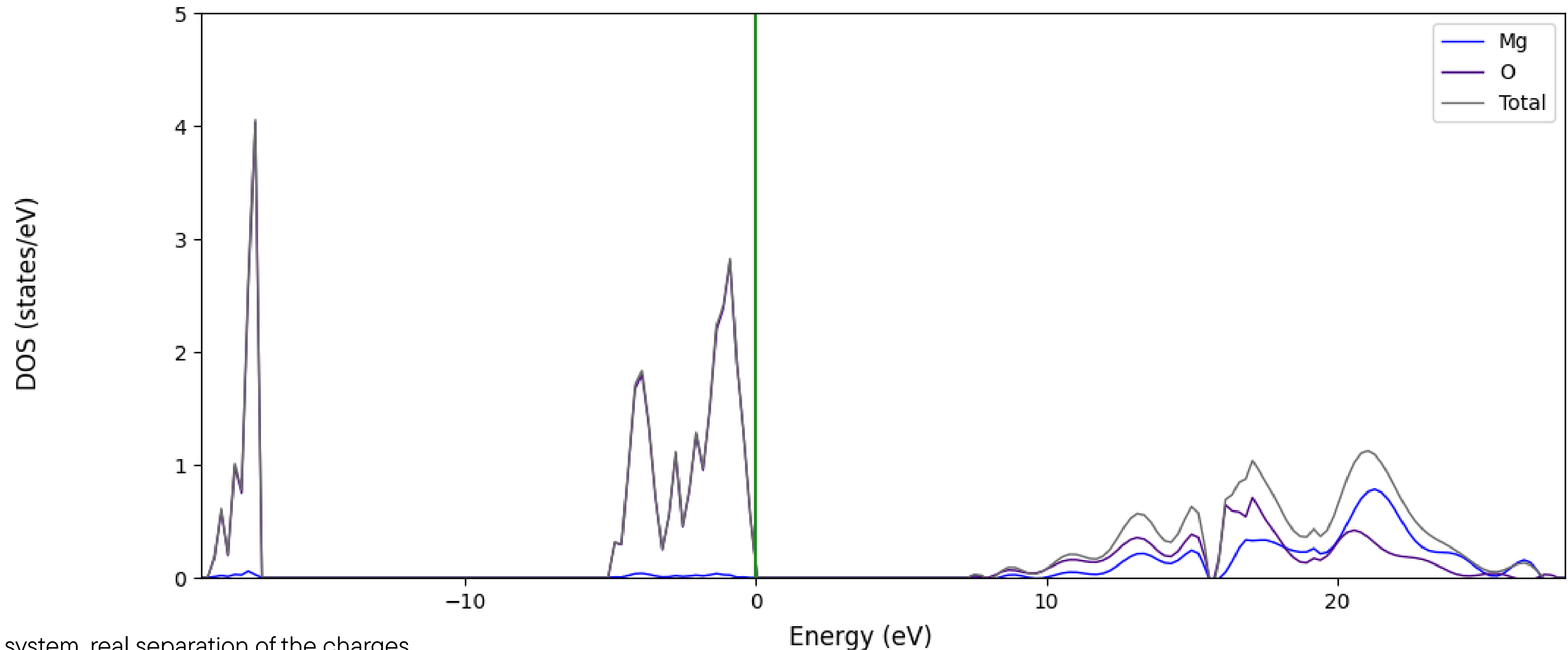
*DOSS computed from eigenvectors so
NEWK keyword is needed

Projection on selected AOs

```
NEWK
8 8
1 0
-----
DOSS                                Keyword
8 200 7 14 1 12 0                Parameters
4 1 2 6 10                        Mg S
3 3 7 11                          Mg PX
3 4 8 12                          Mg PY
3 5 9 13                          Mg PZ
4 14 15 19 23                     O S
3 16 20 24                       O PX
3 17 21 25                       O PY
3 18 22 26                       O PZ
-----
                                AO projection
END
```

Calculation of density of states allows to obtain a chemical insight into the band structure, understanding which atomic species and even which atomic orbitals contribute the most to valence and conduction bands

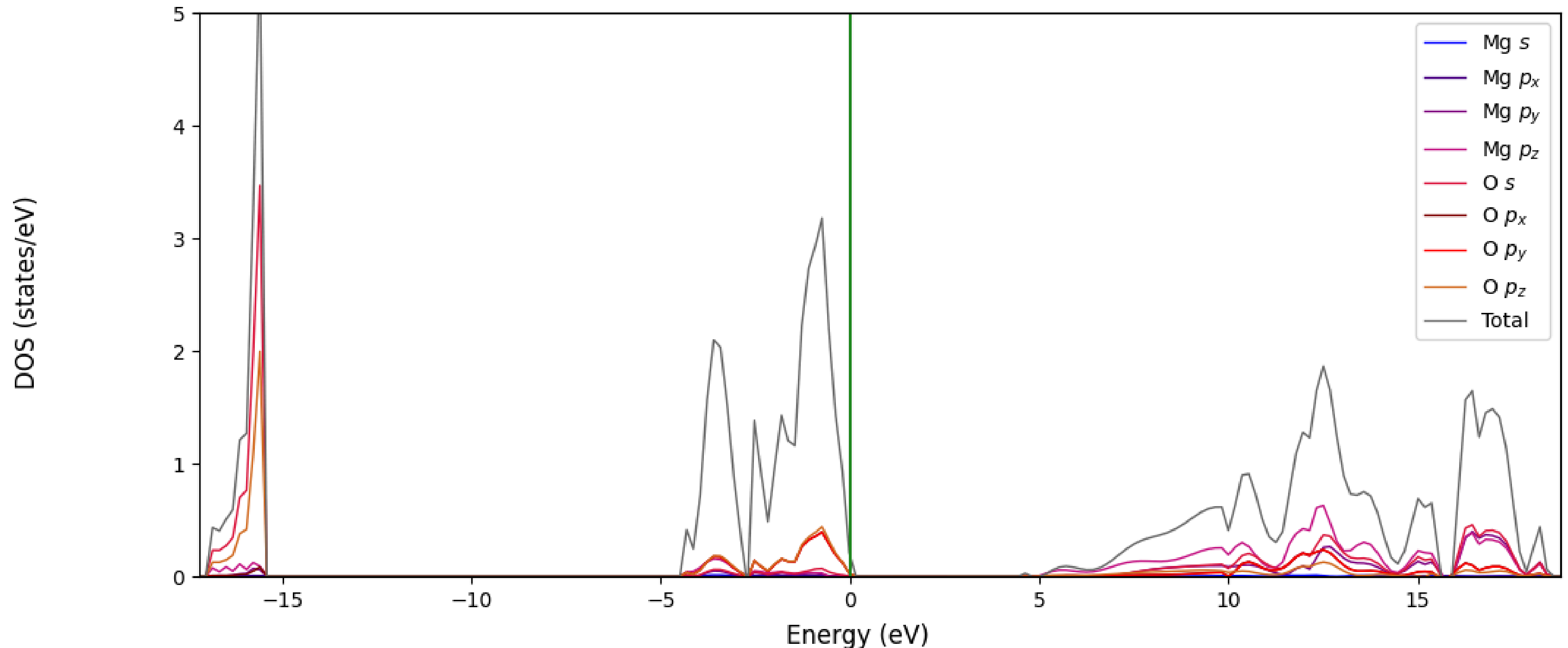
Projection on the total set of AOs of a given atom



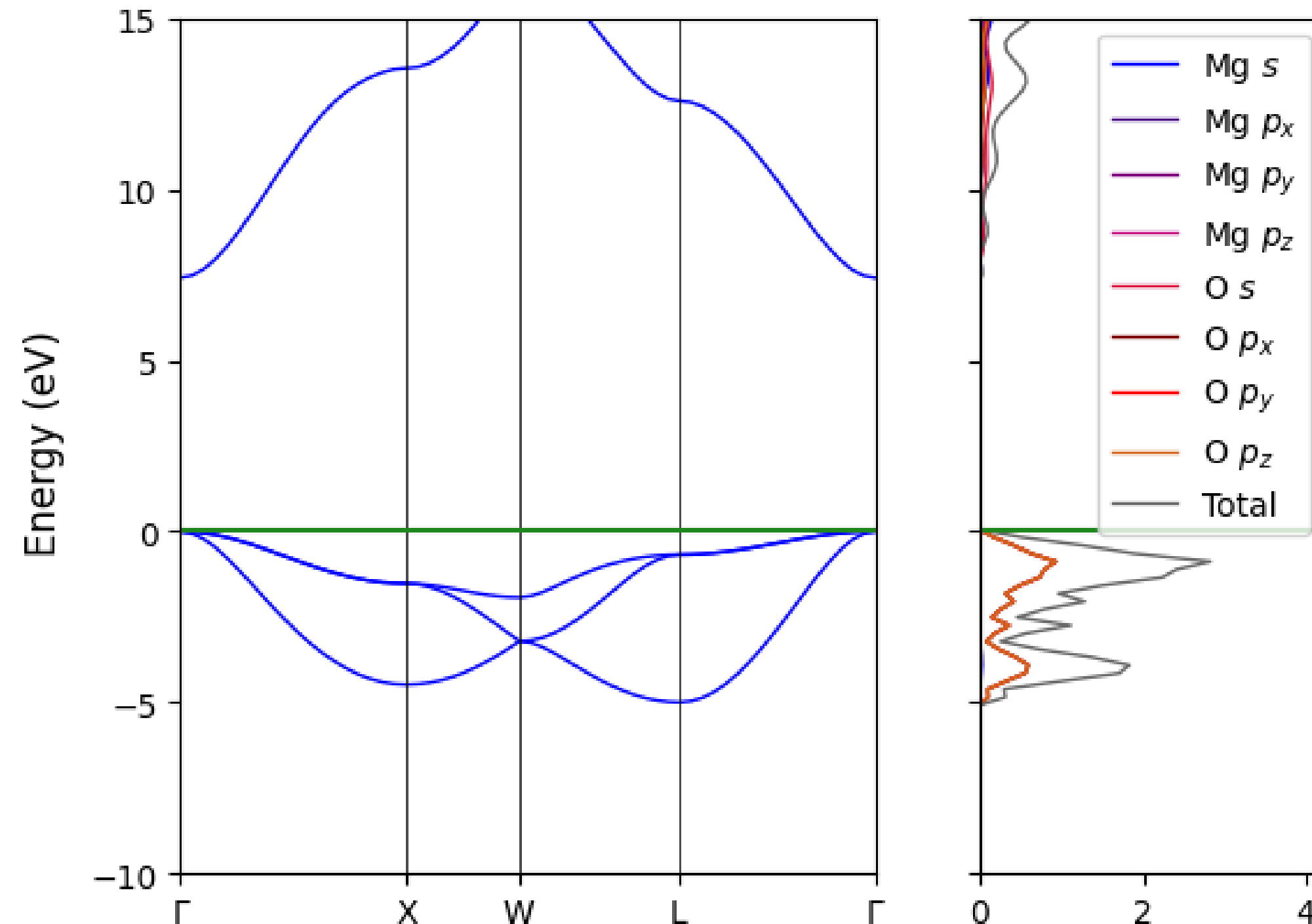
Real ionic system, real separation of the charges

Calculation of density of states allows to obtain a chemical insight into the band structure, understanding which atomic species and even which atomic orbitals contribute the most to valence and conduction bands

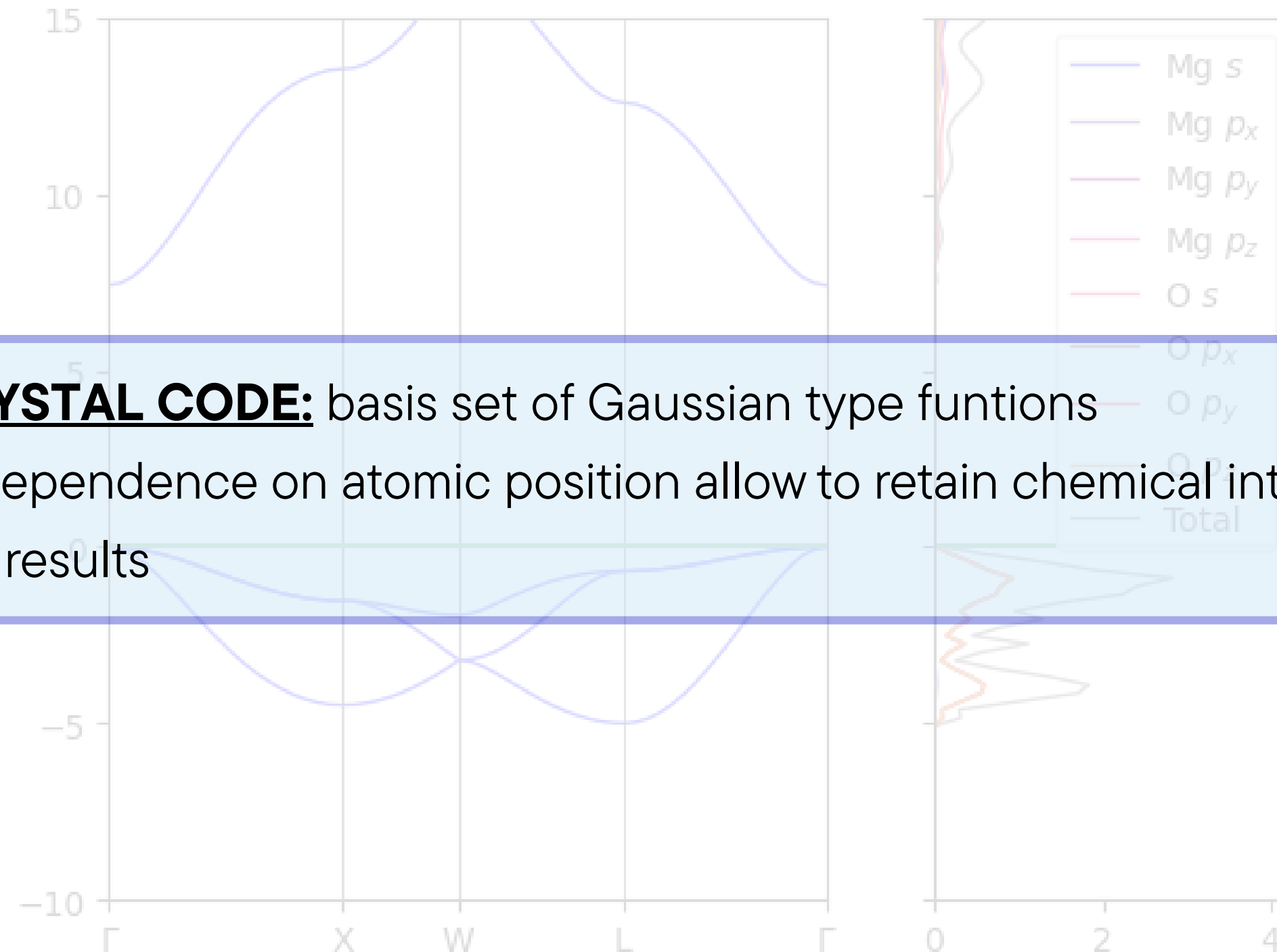
Projection on selected AOs



Calculation of density of states allows to obtain a chemical insight into the band structure, understanding which atomic species and even which atomic orbitals contribute the most to valence and conduction bands



Calculation of density of states allows to obtain a chemical insight into the band structure, understanding which atomic species and even which atomic orbitals contribute the most to valence and conduction bands



STRENGTH OF CRYSTAL CODE: basis set of Gaussian type functions

Local nature, strict dependence on atomic position allow to retain chemical intuition with an easier interpretation of the results

MgO From CRYSTAL wavefunction to *properties* calculation: **ANBD**

ANBD keyword allows to obtain the principal atomic orbital components of selected eigenvectors, obtaining a chemical insight into the band structure of a material, in an even more analytical way respectto DOSS.

Input: *filename.d3*

NEWK	
8 8	
0 0	
ANBD	Keyword
0 0 *	Parameters
0.01	Tolerance
END	
END	

*All k-points, all bands

Input: *filename.d3*

NEWK	
8 8	
0 0	
ANBD	Keyword
1 2	Parameters
0.1	Tolerance
1	* k-point list
10 11	bands list
END	
END	

*Specific bands in Γ point

MgO

From CRYSTAL wavefunction to *properties* calculation: **ANBD**

ANBD keyword allows to obtain the principal atomic orbital components of selected eigenvectors, obtaining a chemical insight into the band structure of a material, in an even more analytical way respect to DOSS.

1

NEWK

8 8

0 0

ANBD

Keyword

0 0

Parameters

0.01

Tolerance

END

END

Output example (Γ point, valence bands)

K POINT 1 (0 0 0)

BAND 7 E(A.U.)= -0.869212

AO 1 S MG 1 RE= 0.02548

AO 2 S MG 1 RE= -0.09615

AO 6 S MG 1 RE= -0.09041

AO 10 S MG 1 RE= 0.08828

AO 14 S O 2 RE= -0.19977

AO 15 S O 2 RE= 0.28106

AO 19 S O 2 RE= 0.43379

AO 23 S O 2 RE= 0.29616

BAND 8 E(A.U.)= -0.176552

AO 3 PX MG 1 RE= 0.07844

AO 7 PX MG 1 RE= 0.01077

AO 16 PX O 2 RE= 0.54134

AO 20 PX O 2 RE= 0.35471

AO 24 PX O 2 RE= 0.45335

BAND 9 E(A.U.)= -0.176552

AO 4 PY MG 1 RE= 0.07844

AO 8 PY MG 1 RE= 0.01077

AO 17 PY O 2 RE= 0.54134

AO 21 PY O 2 RE= 0.35471

AO 25 PY O 2 RE= 0.45335

BAND 10 E(A.U.)= -0.176552

AO 5 PZ MG 1 RE= 0.07844

AO 9 PZ MG 1 RE= 0.01077

AO 18 PZ O 2 RE= 0.54134

AO 22 PZ O 2 RE= 0.35471

AO 26 PZ O 2 RE= 0.45335

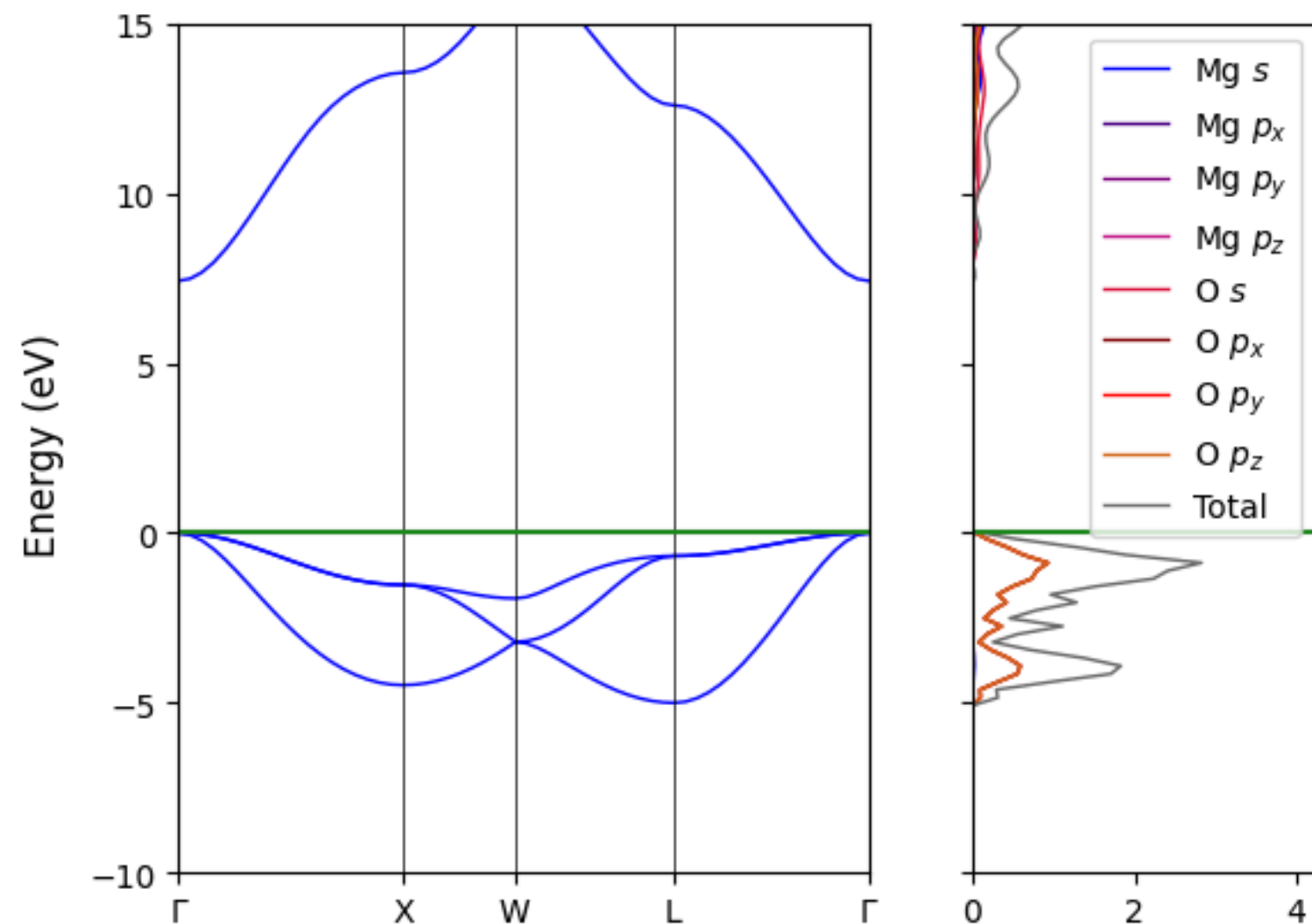
- O orbitals contribute much more to the valence bands respect to Mg (coefficients are one order of magnitude higher)
- The top 3 valence bands (isoenergetic) are related to px, py and pz orbitals of O

MgO

From CRYSTAL wavefunction to *properties* calculation: **ANBD**

ANBD keyword allows to obtain the principal atomic orbital components of selected eigenvectors, obtaining a chemical insight into the band structure of a material, in an even more analytical way respectto DOSS.

①



Output example (Γ point, valence bands)

```
K POINT 1 ( 0 0 0)
BAND 7 E(A.U.)= -0.869212
AO 1 S MG 1 RE= 0.02548
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AO 6 S MG 1 RE= -0.09041
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AO 14 S O 2 RE= -0.19977
AO 15 S O 2 RE= 0.28106
AO 19 S O 2 RE= 0.43379
AO 23 S O 2 RE= 0.29616
BAND 8 E(A.U.)= -0.176552
AO 3 PX MG 1 RE= 0.07844
AO 7 PX MG 1 RE= 0.01077
AO 16 PX O 2 RE= 0.54134
AO 20 PX O 2 RE= 0.35471
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BAND 9 E(A.U.)= -0.176552
AO 4 PY MG 1 RE= 0.07844
AO 8 PY MG 1 RE= 0.01077
AO 17 PY O 2 RE= 0.54134
AO 21 PY O 2 RE= 0.35471
AO 25 PY O 2 RE= 0.45335
BAND 10 E(A.U.)= -0.176552
AO 5 PZ MG 1 RE= 0.07844
AO 9 PZ MG 1 RE= 0.01077
AO 18 PZ O 2 RE= 0.54134
AO 22 PZ O 2 RE= 0.35471
AO 26 PZ O 2 RE= 0.45335
```

- O orbitals contribute much more to the valence bands respect to Mg (coefficients are one order of magnitude higher)
- The top 3 valence bands (isoenergetic) are related to px, py and pz orbitals of O

MgO From CRYSTAL wavefunction to *properties* calculation: **ANBD**

ANBD keyword allows to obtain the principal atomic orbital components of selected eigenvectors, obtaining a chemical insight into the band structure of a material, in an even more analytical way respectto DOSS.

2

NEWK

8 8

0 0

ANBD

Keyword

1 2

Parameters

0.1

Tolerance

1

k-point list

10 11

bands list

END

END

Output example

K POINT 1 (0 0 0)

BAND 10 E(A.U.)= -0.176552

AO 18 PZ O 2 RE= 0.54134

AO 22 PZ O 2 RE= 0.35471

AO 26 PZ O 2 RE= 0.45335

→ **Last valence band**

pz O orbitals just contribute
(considering the threshold we
set)

BAND 11 E(A.U.)= 0.095992

AO 2 S MG 1 RE= 0.26588

AO 6 S MG 1 RE= 0.20784

AO 10 S MG 1 RE= -0.26413

AO 14 S O 2 RE= -0.18637

AO 15 S O 2 RE= 0.15819

AO 19 S O 2 RE= 1.22877

AO 23 S O 2 RE= -1.04183

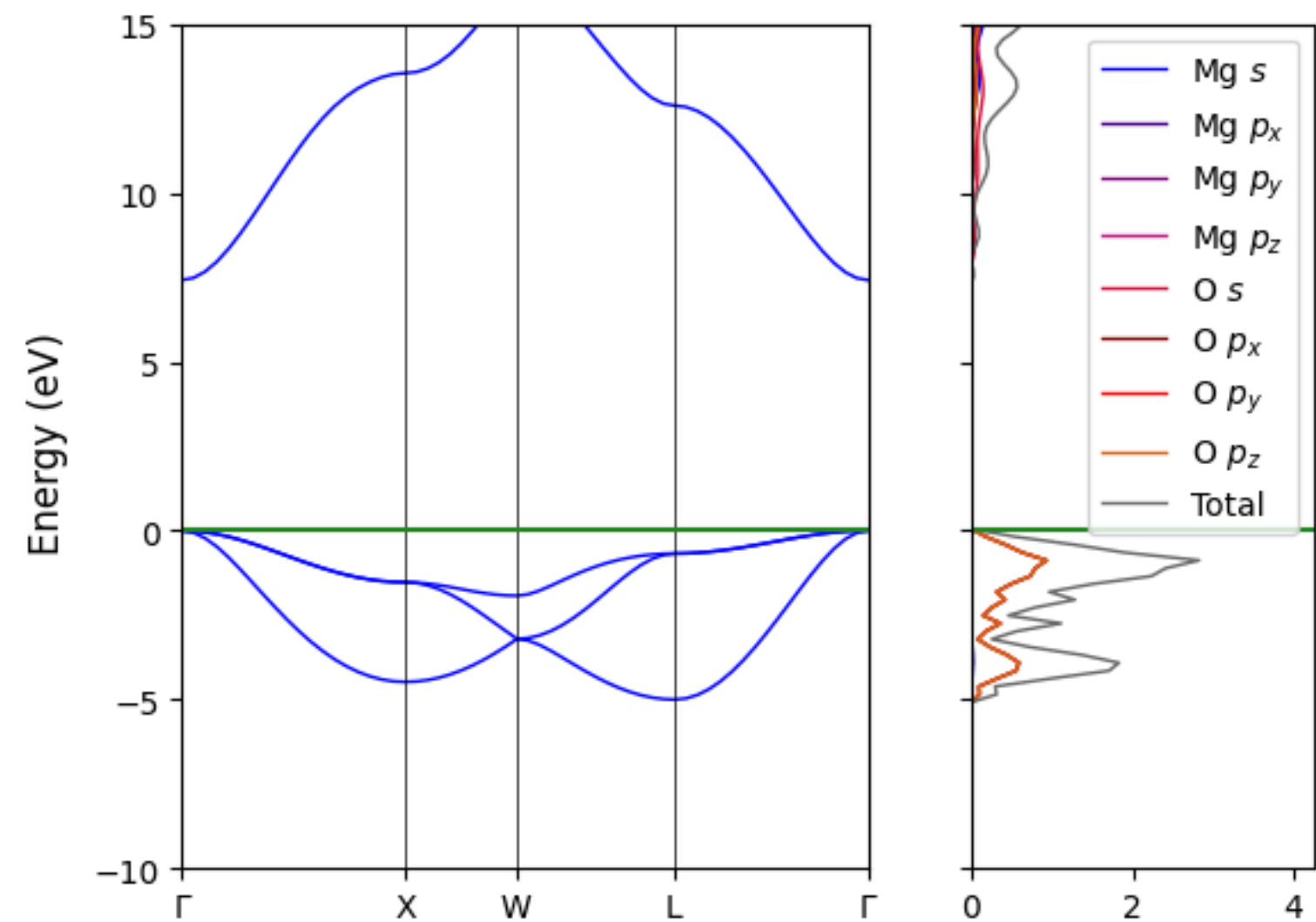
→ **First conduction band**

MgO

ANBD keyword allows to obtain the principal atomic orbital components of selected eigenvectors, obtaining a chemical insight into the band structure of a material, in an even more analytical way respectto DOSS.

2

From CRYSTAL wavefuntion to *properties* calculation: **ANBD**



Same info as DOSS but
here we really have the
eigenvectors
coefficients

Output example

K POINT 1 (0 0 0)				
BAND 10 E(A.U.)= -0.176552				
AO 18	PZ O	2	RE=	0.54134
AO 22	PZ O	2	RE=	0.35471
AO 26	PZ O	2	RE=	0.45335
BAND 11 E(A.U.)= 0.095992				
AO 2	S MG	1	RE=	0.26588
AO 6	S MG	1	RE=	0.20784
AO 10	S MG	1	RE=	-0.26413
AO 14	S O	2	RE=	-0.18637
AO 15	S O	2	RE=	0.15819
AO 19	S O	2	RE=	1.22877
AO 23	S O	2	RE=	-1.04183

MgO From CRYSTAL wavefuntion to *properties* calculation: **PBAN** (or **PDIBAN**).

PBAN keyword allows to compute density matix projected onto a given set of bands.
From the so obtained partial density matrix, properties related to specific bands can be calculated, i.e. access to electronic charge density (ECHG) of some specific bands.

- 1 Electron charge density map of MgO from the whole density matrix

Input: *filename.d3*

ECHG	Keyword
0	Derivatives order
100	N. points
COORDINA	
-4. -4. 0.0	Cartesian coordinates to define the window in a plane
4. -4. 0.0	
4. 4. 0.0	
MARGINS	
1.5 1.5 1.5 1.5	Margins of the window
END	

MgO From CRYSTAL wavefunction to *properties* calculation: **PBAN** (or **PDIBAN**).

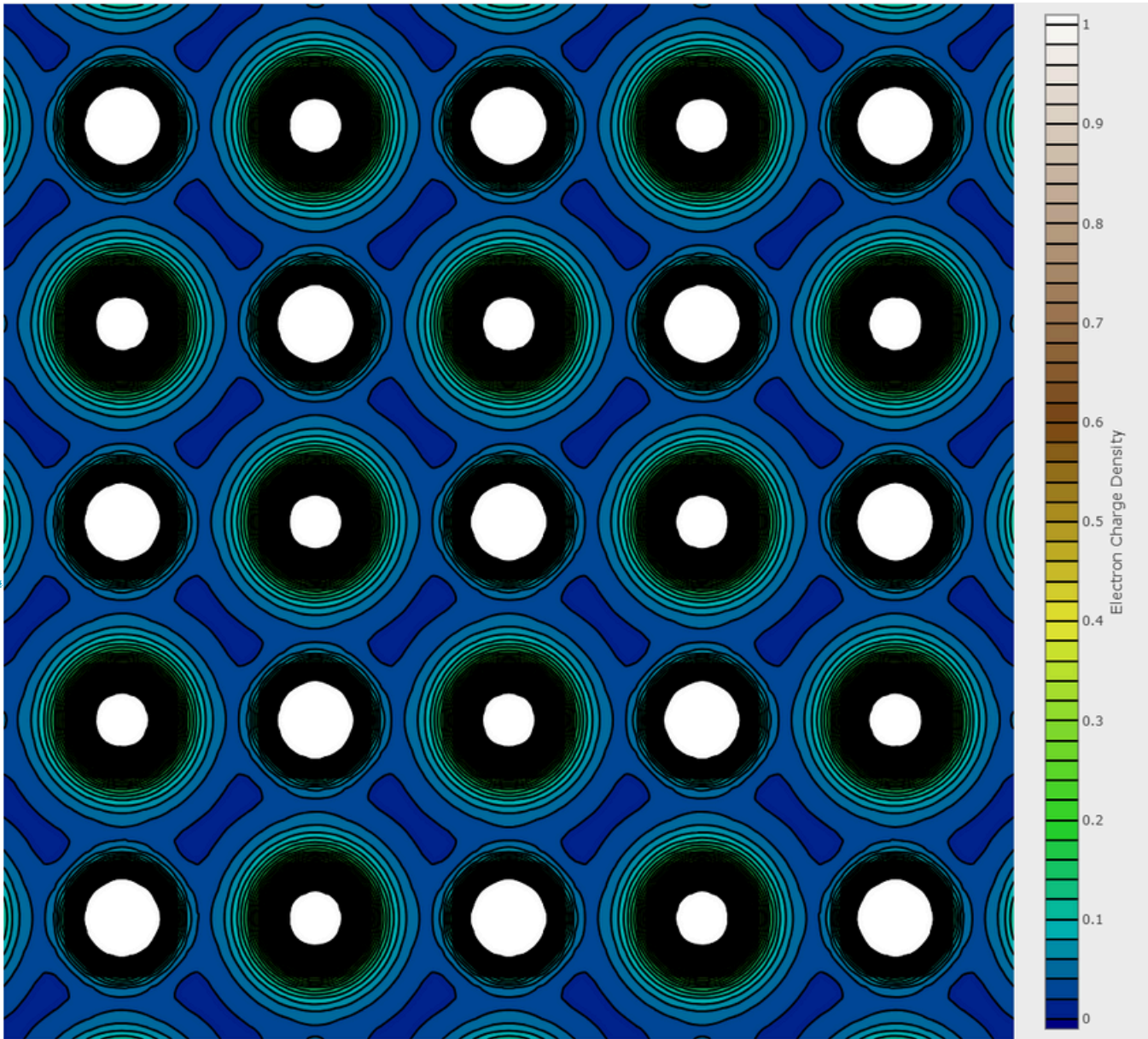
PBAN keyword allows to compute density matrix projected onto a given set of bands.
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Input: *filename.d3*

ECHG	Keyword
0	Derivatives order
100	N. points
COORDINA	
-4. -4. 0.0	Cartesian coordinates to define the window in a plane
4. -4. 0.0	
4. 4. 0.0	
MARGINS	
1.5 1.5 1.5 1.5	Margins of the window
END	

CRYSPLIT



MgO From CRYSTAL wavefunction to *properties* calculation: **PBAN** (or **PDIBAN**).

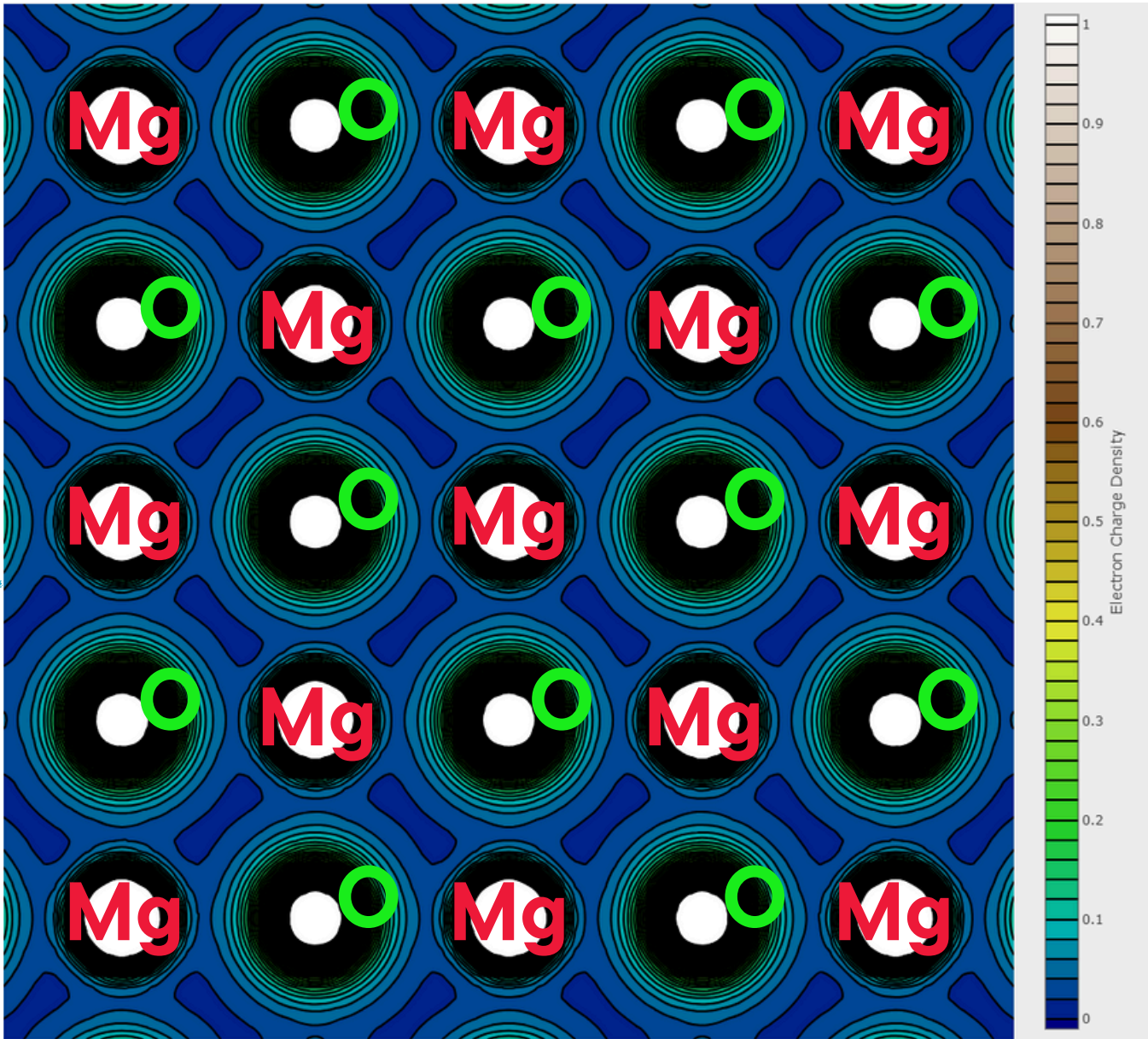
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Input: *filename.d3*

ECHG	Keyword
0	Derivatives order
100	N. points
COORDINA	
-4. -4. 0.0	Cartesian coordinates to define the window in a plane
4. -4. 0.0	
4. 4. 0.0	
MARGINS	
1.5 1.5 1.5 1.5	Margins of the window
END	

CRYSPLOT



MgO From CRYSTAL wavefuntion to *properties* calculation: **PBAN** (or **PDIBAN**).

PBAN keyword allows to compute density matix projected onto a given set of bands.
From the so obtained partial density matrix, properties related to specific can be calculated, i.e. access to electronic charge density (ECHG) of some specific bands.

2 Electron charge density map of MgO from the density matrix projected onto specific valence bands

Input: *filename.d3*

No Symmetry Adapted Bloch Functions	
NOSYMADA	
NEWK	
8 8	
1 0	
PBAN	Keyword
6	N. bands
1 2 3 4 5 6	List
ECHG	
...	Same input as before
END	
END	

No Symmetry Adapted Bloch Functions	
NOSYMADA	
NEWK	
8 8	
1 0	
PBAN	Keyword
4	N. bands
7 8 9 10	List
ECHG	
...	Same input as before
END	
END	

MgO From CRYSTAL wavefunction to *properties* calculation: **PBAN** (or **PDIBAN**).

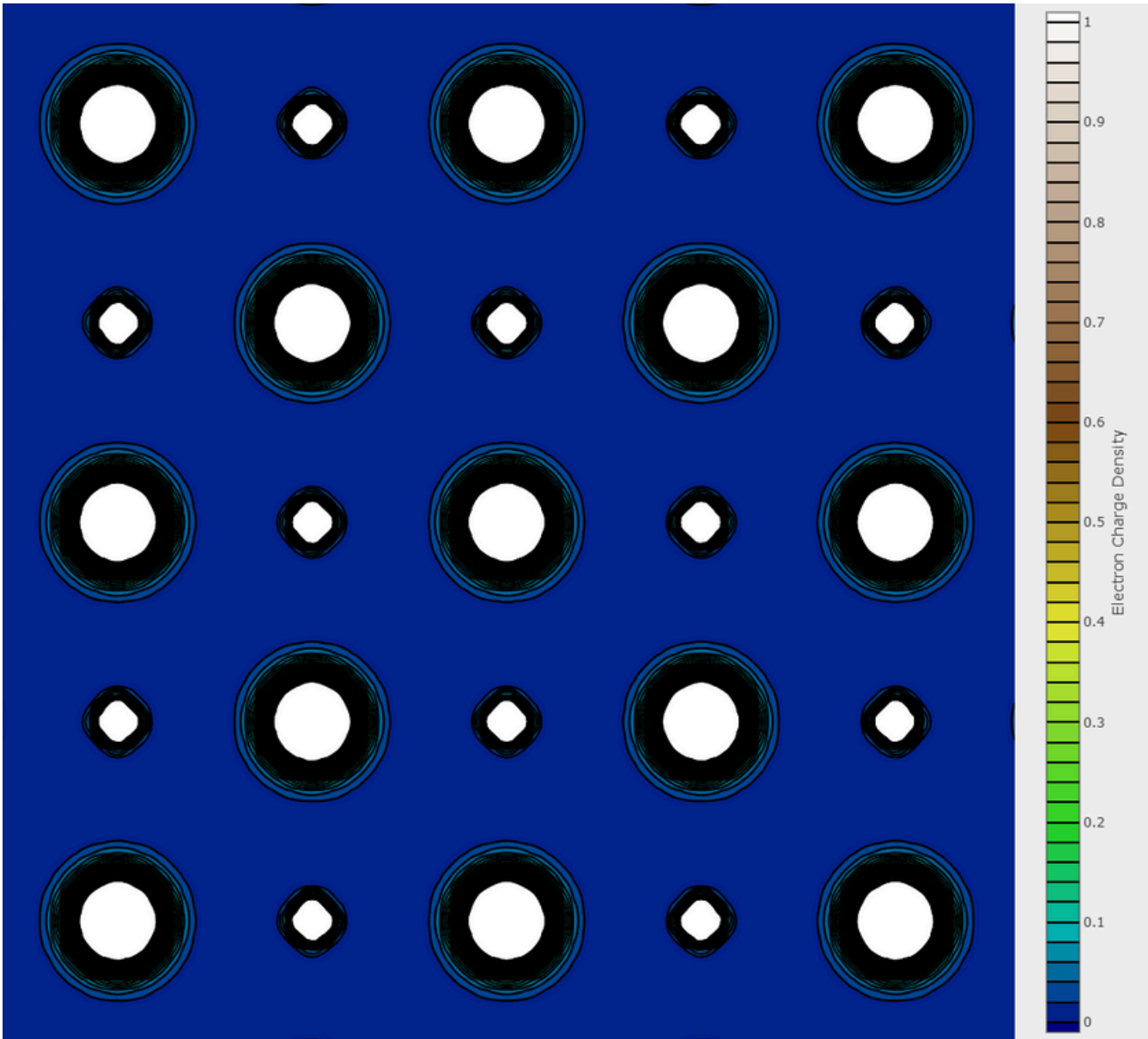
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No Symmetry Adapted Bloch Functions	
NOSYMADA	
NEWK	
8 8	
1 0	
PBAN	Keyword
6	N. bands
1 2 3 4 5 6	List
ECHG	
...	
Same input as before	
END	
END	

Core valence bands

CRYSPLOT



MgO From CRYSTAL wavefunction to *properties* calculation: **PBAN** (or **PDIBAN**).

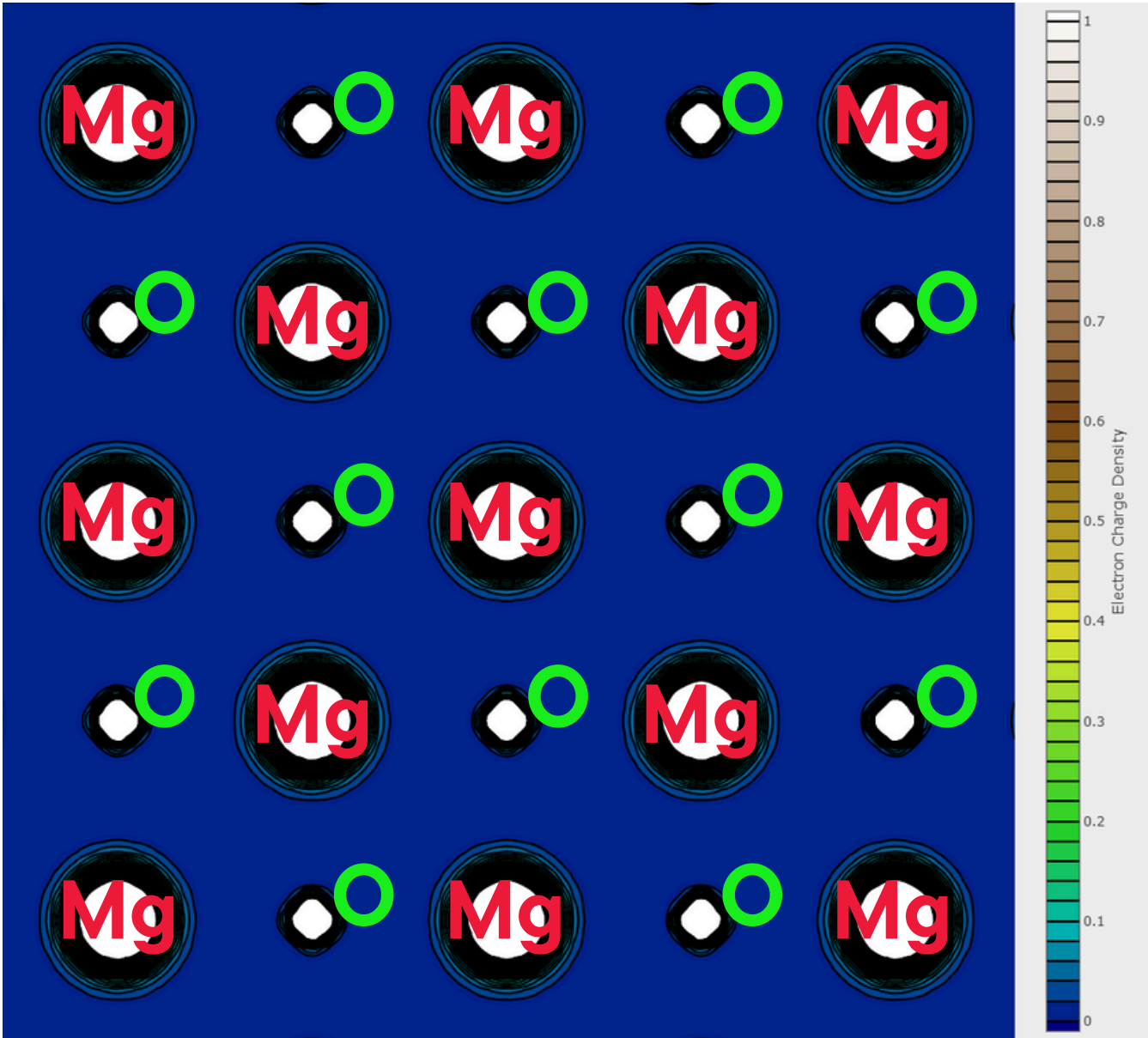
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2 Electron charge density map of MgO from the density matrix projected onto specific valence bands

No Symmetry Adapted Bloch Functions	
NOSYMADA	
NEWK	
8 8	
1 0	
PBAN	Keyword
6	N. bands
1 2 3 4 5 6	List
ECHG	
...	
Same input as before	
END	
END	

Core valence bands

CRYSPLOT



MgO From CRYSTAL wavefunction to *properties* calculation: **PBAN** (or **PDIBAN**).

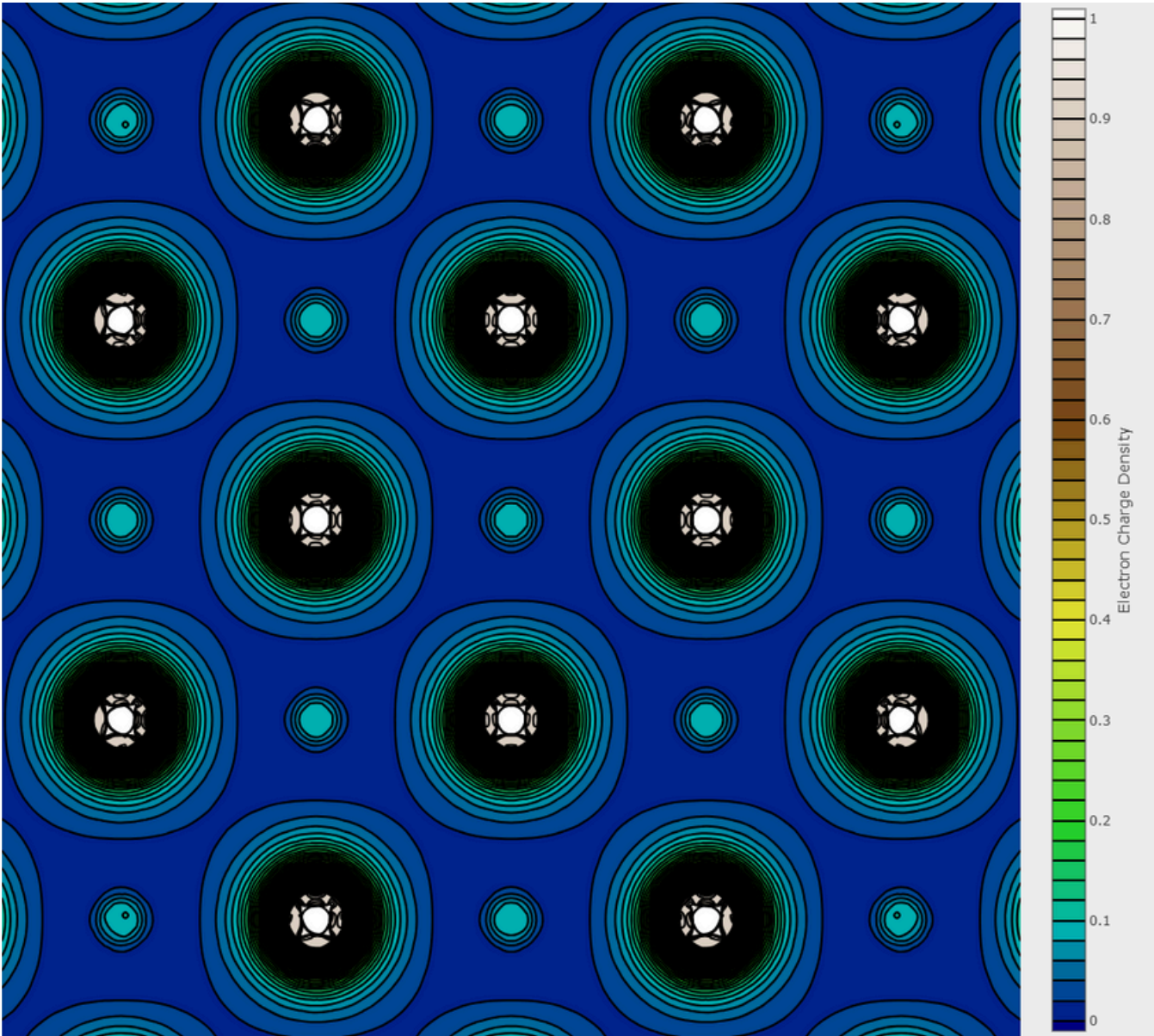
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2 Electron charge density map of MgO from the density matrix projected onto specific valence bands

No Symmetry Adapted Bloch Functions	
NOSYMADA	
NEWK	
8 8	
1 0	
PBAN	Keyword
4	N. bands
7 8 9 10	List
ECHG	
...	
Same input as before	
END	
END	

Top valence bands

CRYSPLIT



MgO From CRYSTAL wavefunction to *properties* calculation: **PBAN** (or **PDIBAN**).

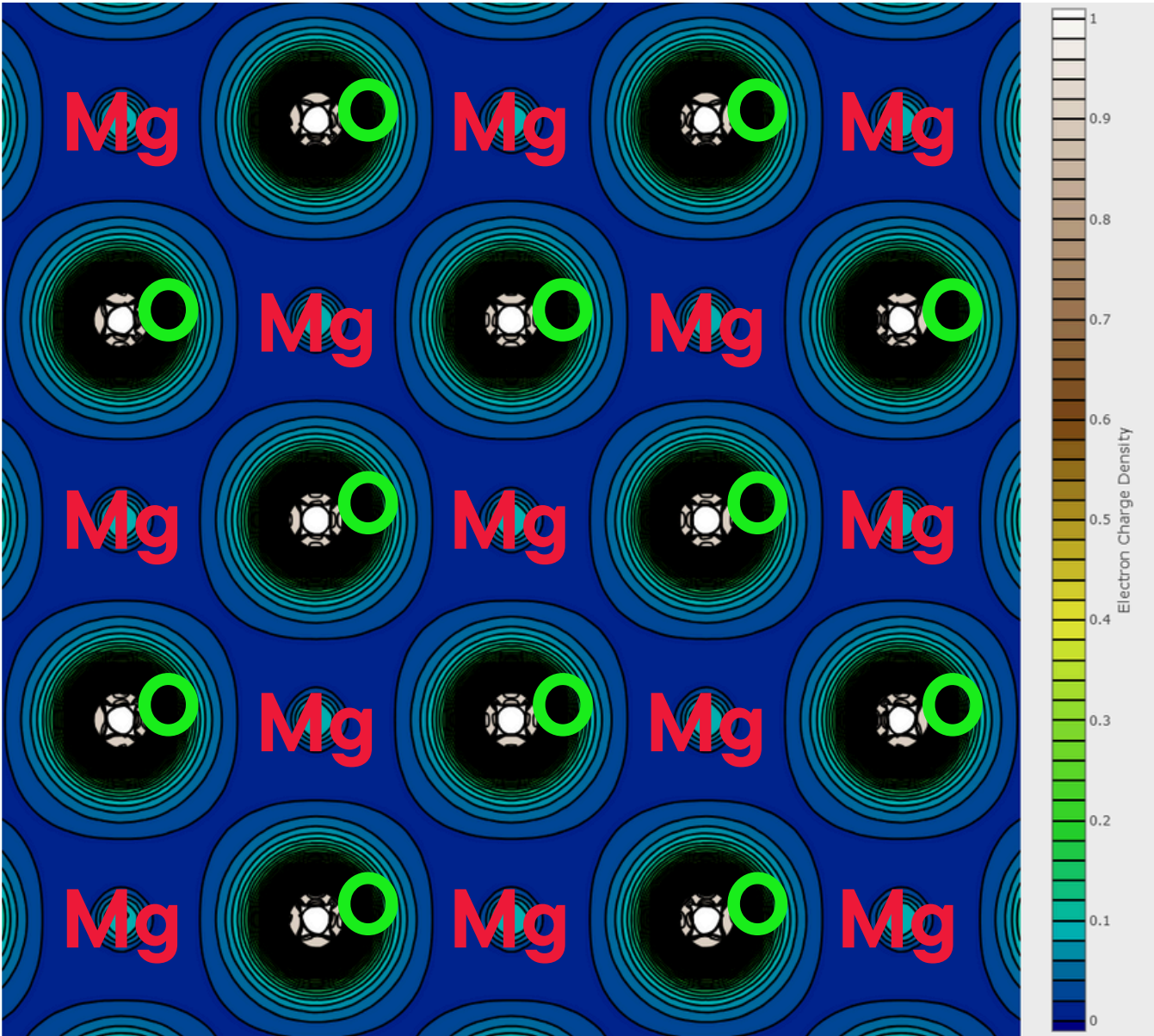
PBAN keyword allows to compute density matrix projected onto a given set of bands.
From the so obtained partial density matrix, properties related to specific can be calculated, i.e. access to electronic charge density (ECHG) of some specific bands.

2 Electron charge density map of MgO from the density matrix projected onto specific valence bands

No Symmetry Adapted Bloch Functions	
NOSYMADA	
NEWK	
8 8	
1 0	
PBAN	Keyword
4	N. bands
7 8 9 10	List
ECHG	
...	Same input as before
END	
END	

Top valence bands

CRYSPLOT



MgO From CRYSTAL wavefunction to *properties* calculation: **PDIDE**

Similarly to PBAN, PDIDE keyword allows to compute density matrix projected onto a given energy range from which it is possible to compute properties related to specific bands in that range, i.e charge density maps obtained give the STM topography within the Tersoff-Haman approximation.

Input: *filename.d3*

```
NEWK
8 8
1 0
-----
PDIDE                                Keyword
-1 -0.1                             Energy range (Ha)
-----
ECHG
...                                  Same input as before
-----
END
END
```


MgO From CRYSTAL wavefunction to *properties* calculation: PDIDE

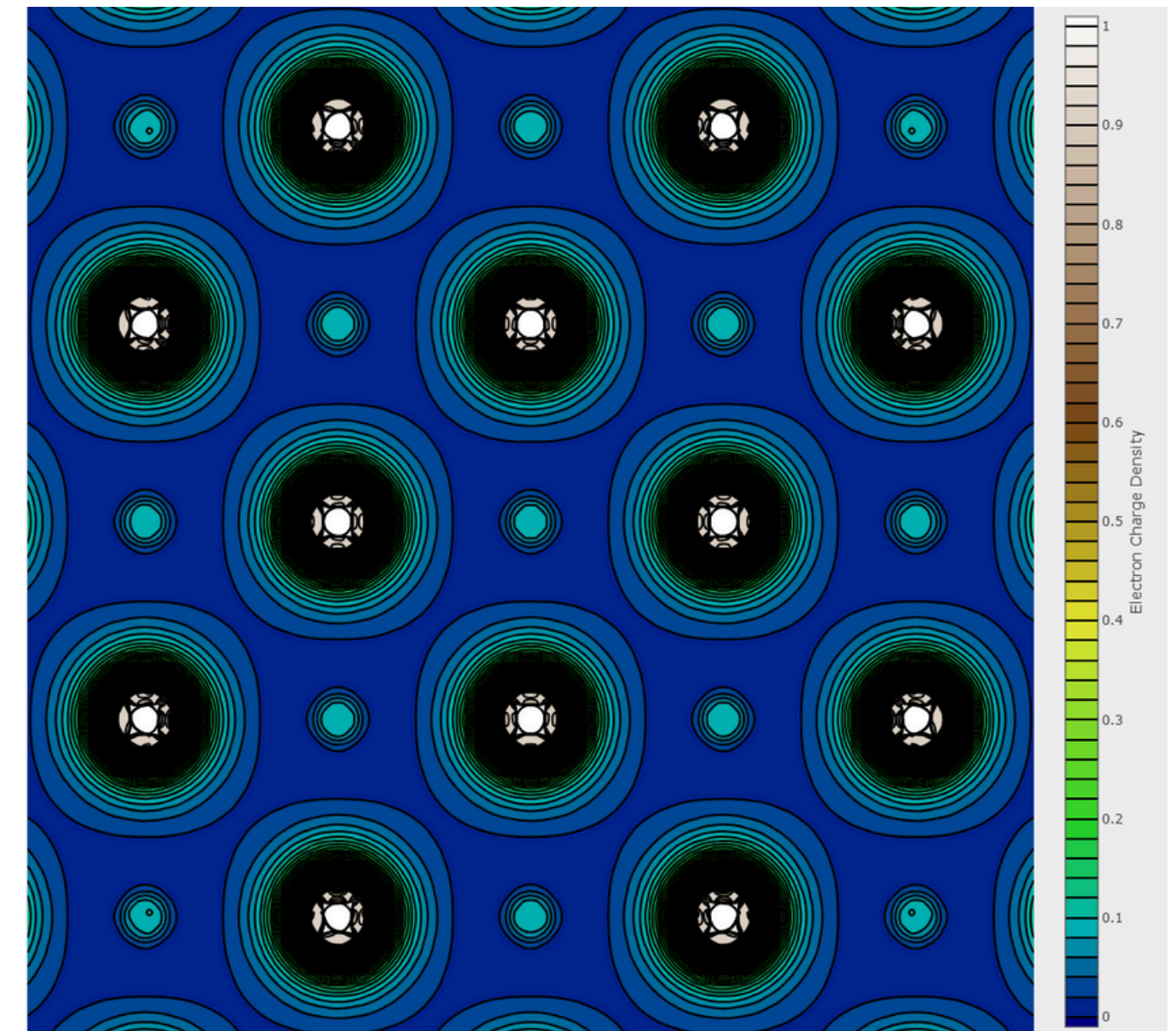
Similarly to PBAN, PDIDE keyword allows to compute density matrix projected onto a given energy range from which it is possible to compute properties related to specific bands in that range, i.e charge density maps obtained give the STM topography within the Tersoff-Haman approximation.

Input: *filename.d3*

NEWK	
8 8	
1 0	
PDIDE	Keyword
-1 -0.1*	Energy range (Ha)
ECHG	
...	Same input as before
END	
END	

*Top valence bands
(from ANBD output)

CRYSPLOT



MgO From CRYSTAL wavefunction to *properties* calculation: PDIDE

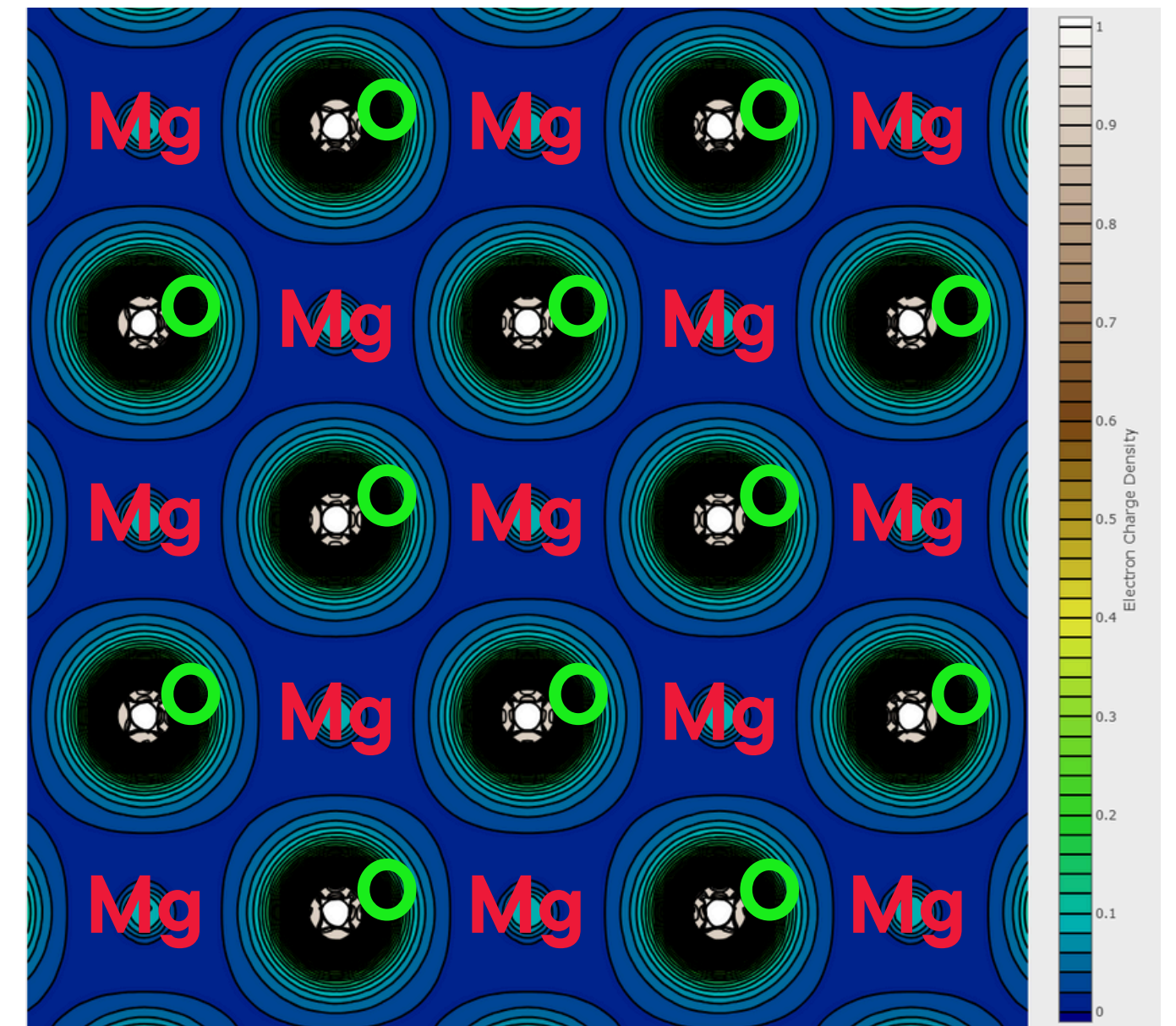
Similarly to PBAN, PDIDE keyword allows to compute density matrix projected onto a given energy range from which it is possible to compute properties related to specific bands in that range, i.e charge density maps obtained give the STM topography within the Tersoff-Haman approximation.

Input: *filename.d3*

NEWK	
8 8	
1 0	
PDIDE	Keyword
-1 -0.1	Energy range (Ha)
ECHG	
...	Same input as before
END	
END	

Top valence bands

CRYSPLOT



MgO From CRYSTAL wavefunction to *properties* calculation: **ECH3** and **POT3**

ECH3 → electronic charge density (electron/bohr) at a regular 3-dimensional grid of points

POT3 → electrostatic potential at a regular 3-dimensional grid of points

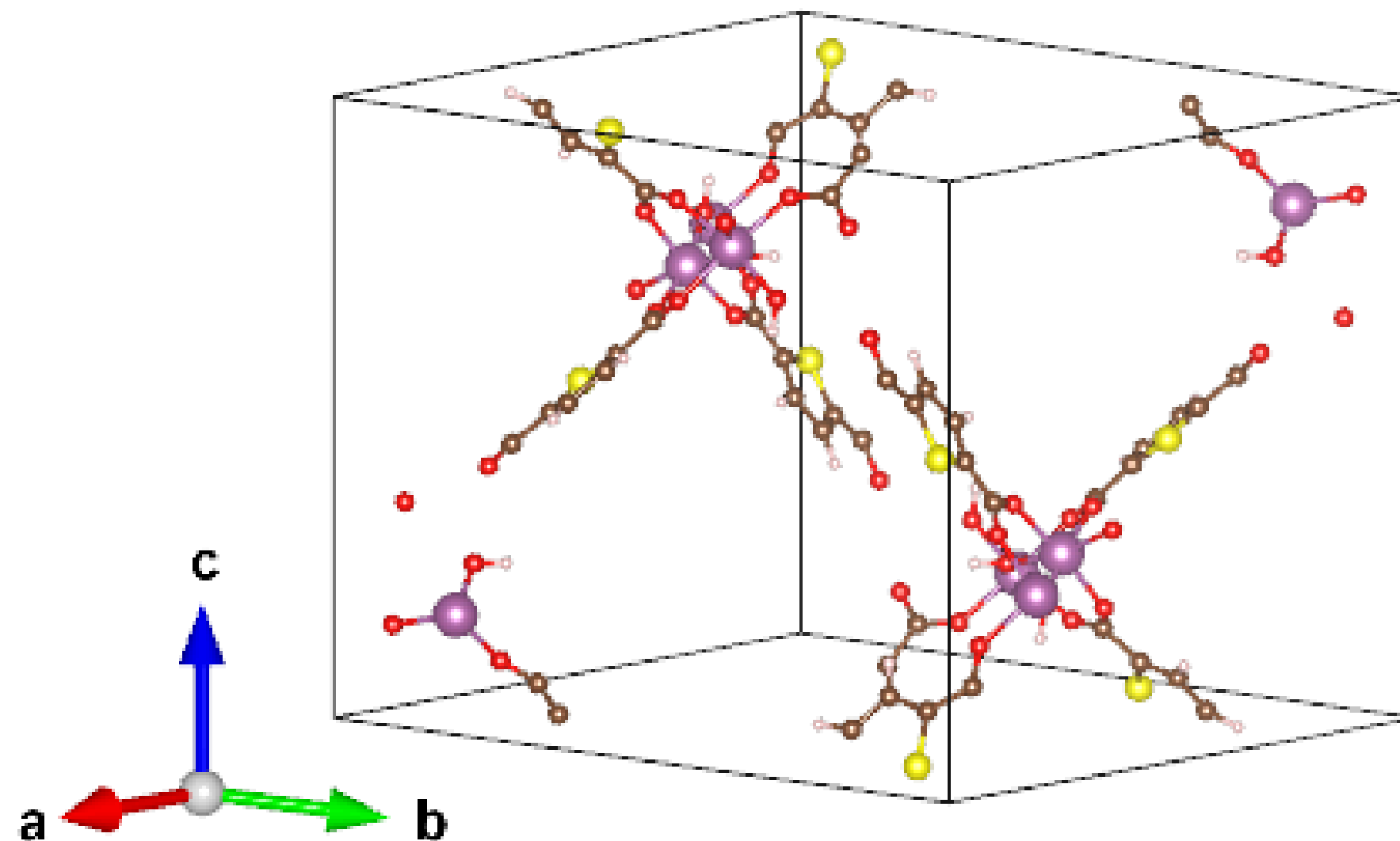
ECH3	Keyword
100	N. sampling points in one direction
POT3	Keyword
100	N. sampling points in one direction
5	Penetration tolerance
END	

DENS_CUBE.DAT

POT_CUBE.DAT

Metal-organic frameworks (MOFs)

NOTT-401

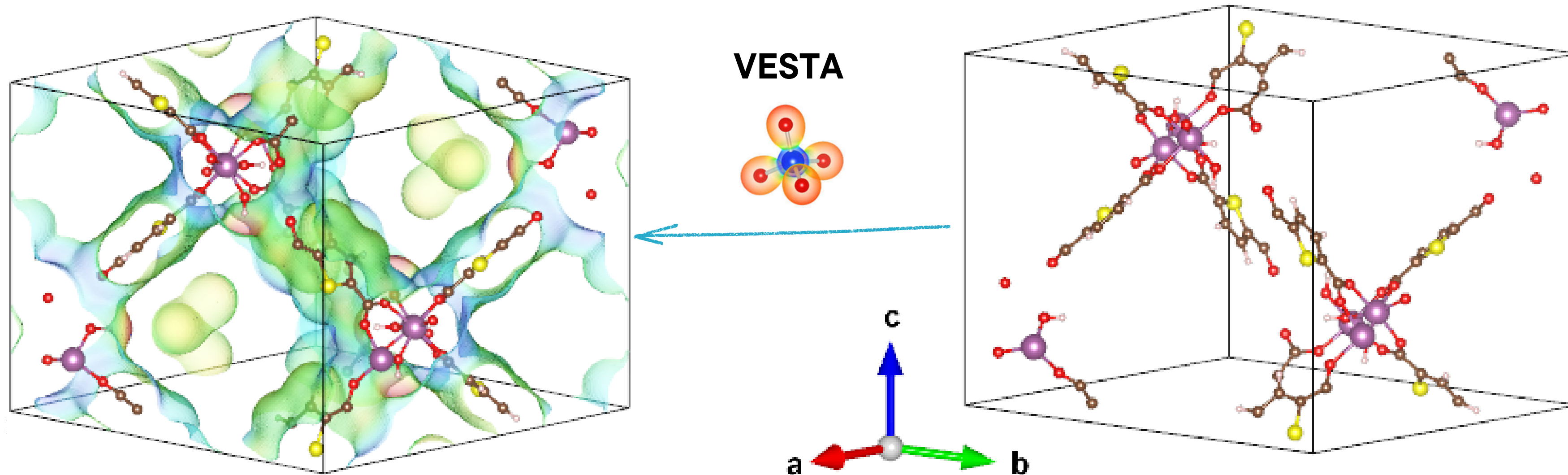


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Metal-organic frameworks (MOFs)
NOTT-401



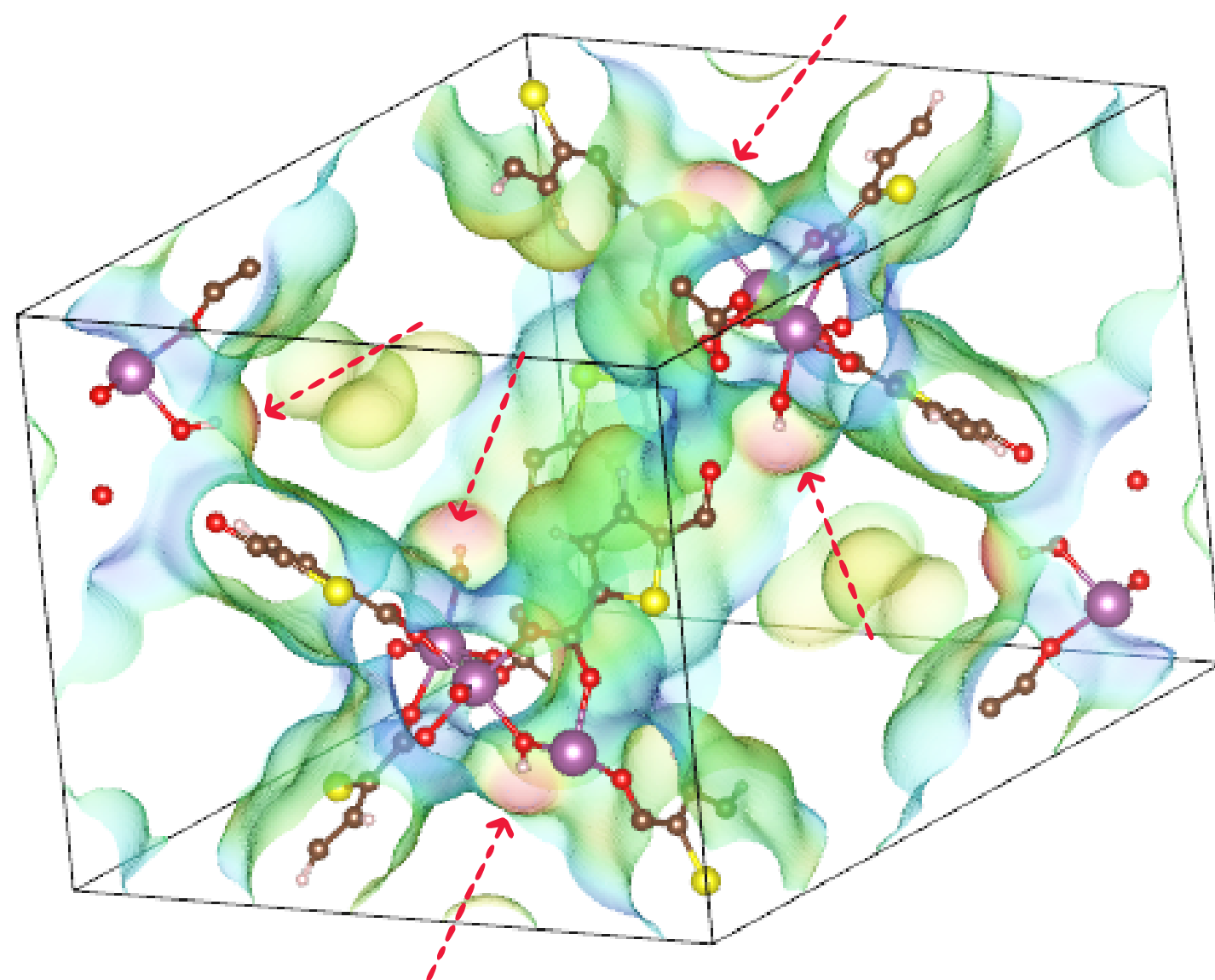
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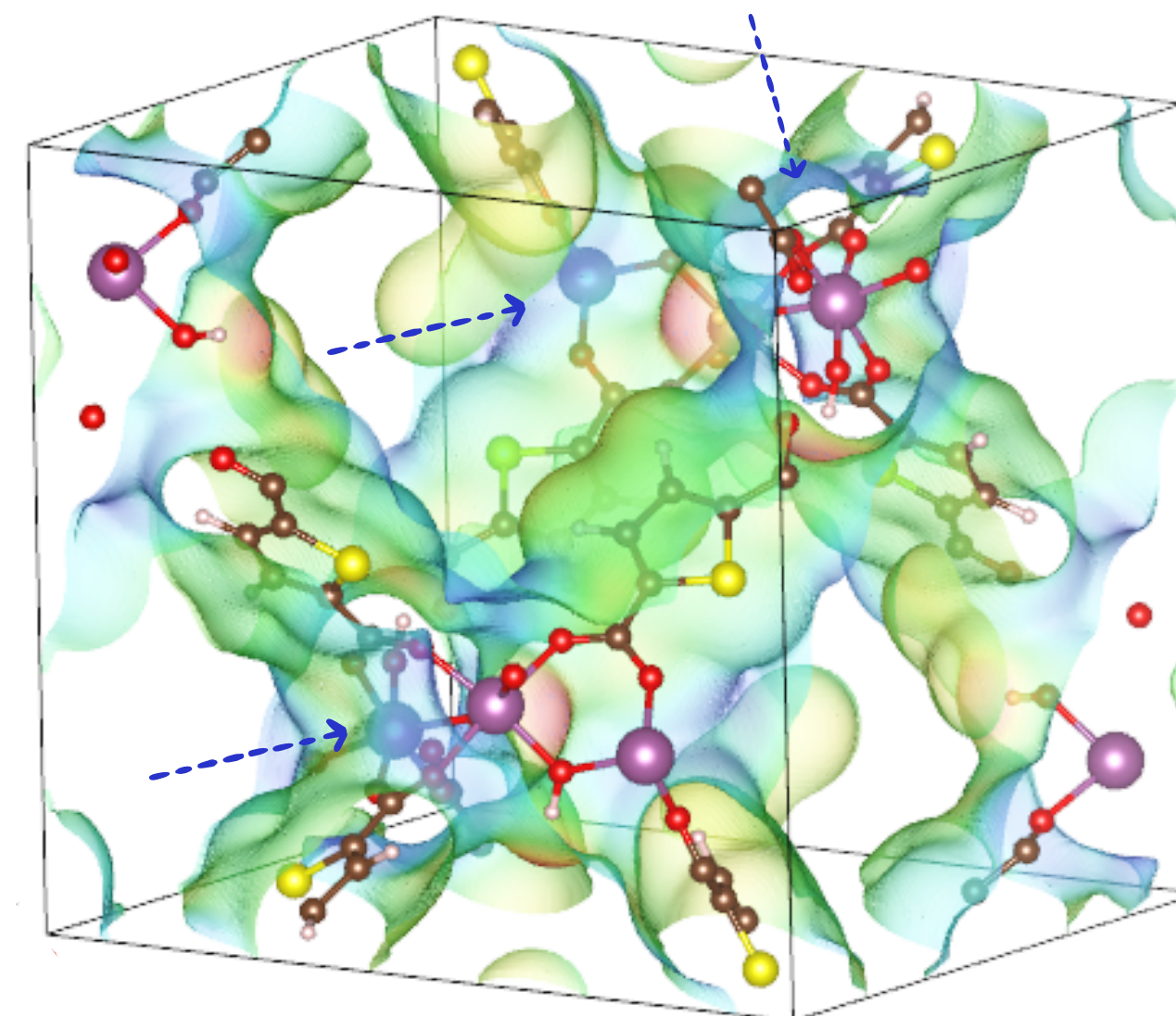
POT3 → electrostatic potential at a regular 3-dimensional grid of points

Identify preferential adsorption sites

Red zones: electron poor, positively “charged”



Blue zones: electron rich, negatively “charged”



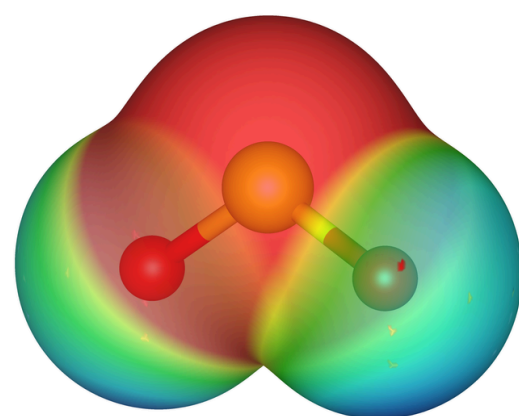
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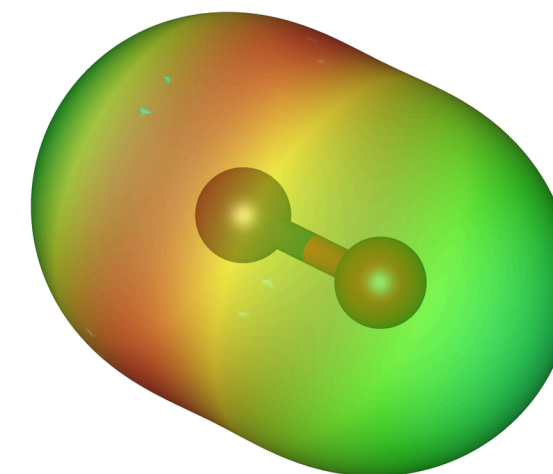
Identify preferential adsorption sites

SO₂ molecule



- S atom from the SO₂ molecule is interacting with three oxygen atoms provided by NOTT-401
- O atoms are acceptors of H bonds → O–H...O and C–H...O

CO molecule



- Weak interactions with the MOF